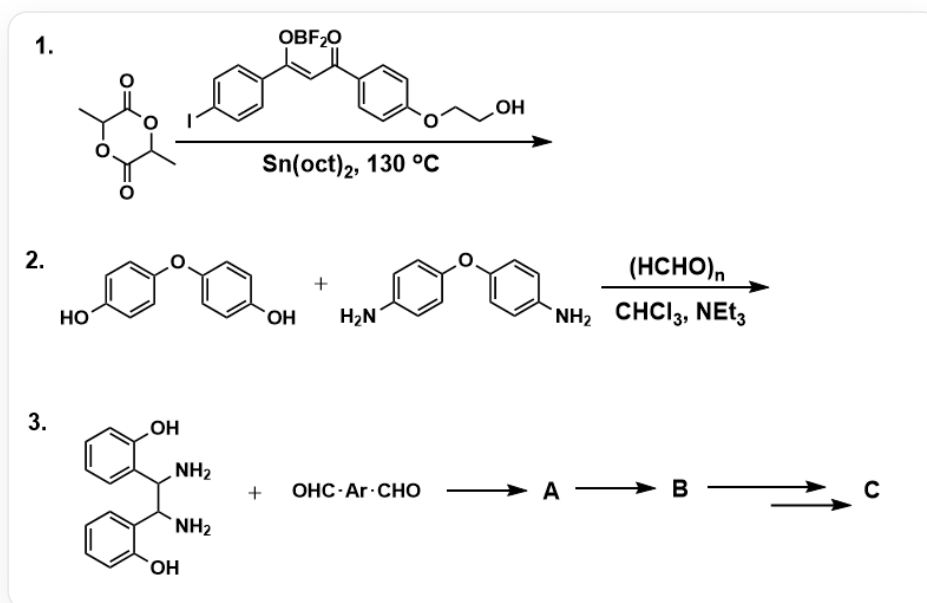


Question

For the following polymerization reactions, the following information is available:



There are three chemical reactions in the figure. 1 is

O=C(C(C)O1)OC(C)C1=O>IC2=CC=C(/C(=O)OC3=CC=C(C=C3)OC(=O)C=C2>, condition is $\text{Sn(oct)}_2, 130^\circ\text{C}$; 2 is

OC1=CC=C(OC2=CC=C(O)C=C2)C=C1.NC3=CC=C(OC4=CC=C(N)C=C4)C=C3>>, condition is

Extra close brace or missing open brace; 3 is OC1=CC=CC=C1C(C(C2=CC=CC=C2O)N)N.O=C[Ar]C=O>>[*],

subsequently product **A** in one step generates **B** then in multiple steps generates **C**

1. In reaction 2, the two substrates and formaldehyde react in a 1:1:4 ratio to obtain a chain polymer. The ^1H NMR of the product shows that the structure does not contain active hydrogen. Note: This ratio is calculated based on formaldehyde monomers.

2. Reaction 3 obtains a polymer with a classic salen ligand as a side group through condensation and pericyclic reactions.

Which of the following statements is incorrect (select A if all are correct)?

A. All other options are incorrect

- B.** In reaction 1, one repeating unit of the polymer product contains 9 atoms.
- C.** Reaction 1 is an addition polymerization reaction.
- D.** In reaction 2, one repeating unit of the polymeric product contains 6 rings.
- E.** In reaction 3, the pericyclic reaction is specifically a $[3,3]\sigma$ sigmatropic rearrangement.
- F.** Reaction 3 is a condensation reaction.

Answer

Correct Answer: **A**

Detailed Explanation

In the following analysis, when the polymer repeating unit is represented using SMILES, R1 and R2 represent the end groups of the polymer repeating unit.

For reaction 1, IC2=CC=C(/C(OB(F)F)=C/C(C3=CC=C(OCCO)C=C3)=O)C=C2 is the initiator, acting as an end group, initiating the addition polymerization of O=C(C(C)O1)OC(C)C1=O, the repeating unit in the product is [R1]OC(C([R2]))C=O, and the end groups R1 and R2 are hydroxyl and IC1=CC=C(/C(OB(F)F)=C/C(C2=CC=C(OCCO[R])C=C2)=O)C=C1, respectively. Therefore, the repeating unit contains 9 atoms, B is correct; addition polymerization, C is correct.

CHECKPOINT

1 PTS

The repeating unit in the product of reaction 1 is [R1]OC(C([R2]))C=O

CHECKPOINT

1 PTS

The end groups R1 and R2 are hydroxyl and IC1=CC=C(/C(OB(F)F)=C/C(C2=CC=C(OCCO[R])C=C2)=O)C=C1, respectively

For reaction 2, since the reaction ratio is 1:1:4, it can be determined that 1 molecule of substrate will react with 4 molecules of formaldehyde, undergoing 4 steps of Mannich reaction, forming a polymer in which the repeating unit

is CC1=CC=C(OC2=CC=C(N3CC4=CC(OC5=CC6=C(OCN(C6)C)C=C5)=CC=C4OC3)C=C2)C=C1, containing 6 rings, D is correct.

CHECKPOINT

1 PTS

The repeating unit of the product of reaction 2 is CC1=CC=C(OC2=CC=C(N3CC4=CC(OC5=CC6=C(OCN(C6)C)C=C5)=CC=C4OC3)C=C2)C=C1

For reaction 3, condensation and pericyclic reactions occurred according to the prompt. **A** is the condensation product OC1=C(C(/N=C/[Ar]C=O)C(/N=C/[Ar]C=O)C2=C(O)C=CC=C2)C=CC=C1.

CHECKPOINT

1 PTS

A is OC1=C(C(/N=C/[Ar]C=O)C(/N=C/[Ar]C=O)C2=C(O)C=CC=C2)C=CC=C1

A → **B** is a pericyclic reaction, at this time cope rearrangement occurs in the structure of **A**, which can form a salen ligand analog, E is correct. **B** is OC1=C(/C=N/C([Ar]C=O)C(/N=C/C2=CC=CC=C2O)[Ar]C=O)C=CC=C1.

CHECKPOINT

1 PTS

Cope rearrangement occurs in the structure of **A**

CHECKPOINT

1 PTS

B is OC1=C(/C=N/C([Ar]C=O)C(/N=C/C2=CC=CC=C2O)[Ar]C=O)C=CC=C1

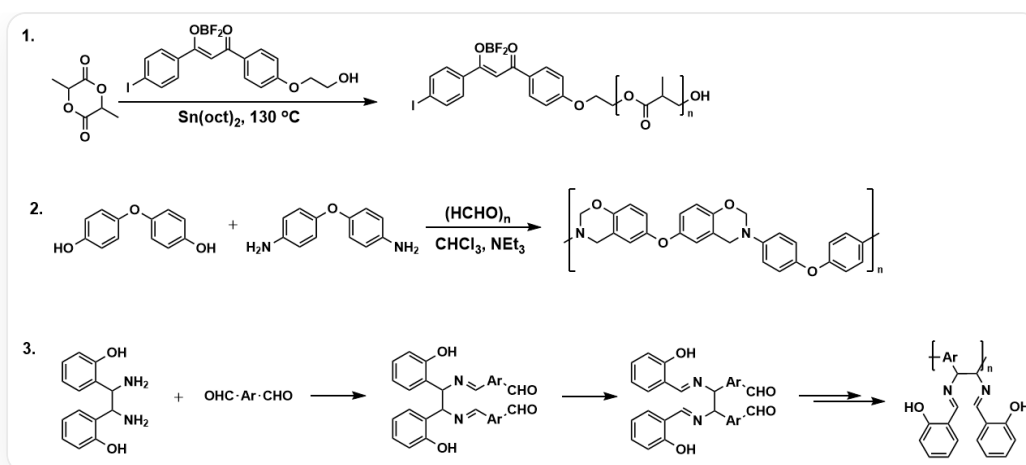
Subsequently, **B** → **C** undergoes aldol condensation polymerization, which is a condensation reaction, F is correct. The repeating unit of the product is [R2][Ar]C(/N=C/C1=C(O)C=CC=C1)C(/N=C/C2=C(O)C=CC=C2)[R1].

CHECKPOINT

1 PTS

The repeating unit of the product of reaction 3 is [R2][Ar]C(/N=C/C1=C(O)C=CC=C1)C(/N=C/C2=C(O)C=CC=C2)[R1]

Therefore, all statements are correct, choose A.



The three reaction processes and products are shown in this figure. The repeating unit in the product of reaction 1 is $[R1]OC(C([R2])C=O)$, and the end groups R1 and R2 are hydroxyl and $IC1=CC=C(/C(OB(F)F)=C/C(C2=CC=C(OCCO[R])C=C2)=O)C=C1$, respectively; the repeating unit of the product of reaction 2 is $CC1=CC=C(OC2=CC=C(N3CC4=CC(OC5=CC6=C(OCN(C6)C)C=C5)=CC=C4OC3)C=C2)C=C1$; in reaction 3, **A** is $OC1=C(C(/N=C/[Ar]C=O)C(/N=C/[Ar]C=O)C2=C(O)C=CC=C2)C=CC=C1$; **B** is $OC1=C(/C=N/C([Ar]C=O)C(/N=C/C2=CC=CC=C2O)[Ar]C=O)C=CC=C1$; the repeating unit of the product is $[R2][Ar]C(/N=C/C1=C(O)C=CC=C1)C(/N=C/C2=C(O)C=CC=C2)[R1]$